

Knowledge Representation

Frank van Harmelen

Who am I?

- Frank van Harmelen
 - <http://www.cs.vu.nl/~frank.van.harmelen>
 - Studied Maths & CS in Amsterdam
 - PhD in AI from Edinburgh
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- Prof at VU, co-leading a group of ± 15 researchers
 - Knowledge representation, Semantic Web, Knowledge Graphs, Medical applications

Examples of intelligent behaviour?

carry out complex reasoning
(solve physics problems,
prove theorems)

draw plans

(e.g.

use natural

/

“but we do not mean that people can recognize familiar objects,
or execute motor skills, or navigate around space,
abilities we share with dogs and cats (and fish)”

conversations)

solving novel complex problems
(generating plans,
designing artifacts)

But isn't modern AI all about Machine Learning ?

Two main lines of development in AI

- symbolic representations
- statistical representation

There have been alternating cycles of one dominating over the other in different decades of the history of AI.

A stairway?

Connectionist
Data
Statistics
Learning

Symbolic
Knowledge
Logic
Reasoning

A pendulum!

Connectionist
Data
Statistics
Learning

Symbolic
Knowledge
Logic
Reasoning

A GOOD JOB.

Statistical vs. symbolic AI: very different types of applications

statistical:

- pattern recognition (images, sound, shapes)
- motor skills (robots)
- speech generation (sound)
- search engines

symbolic:

- planning (autonomous space missions)
- reasoning (diagnosis, design, decision support)
- language generation (conversations)
- search engines

**Who was the previous
president of the United States?**

**Who was the previous previous
president of the United States?**

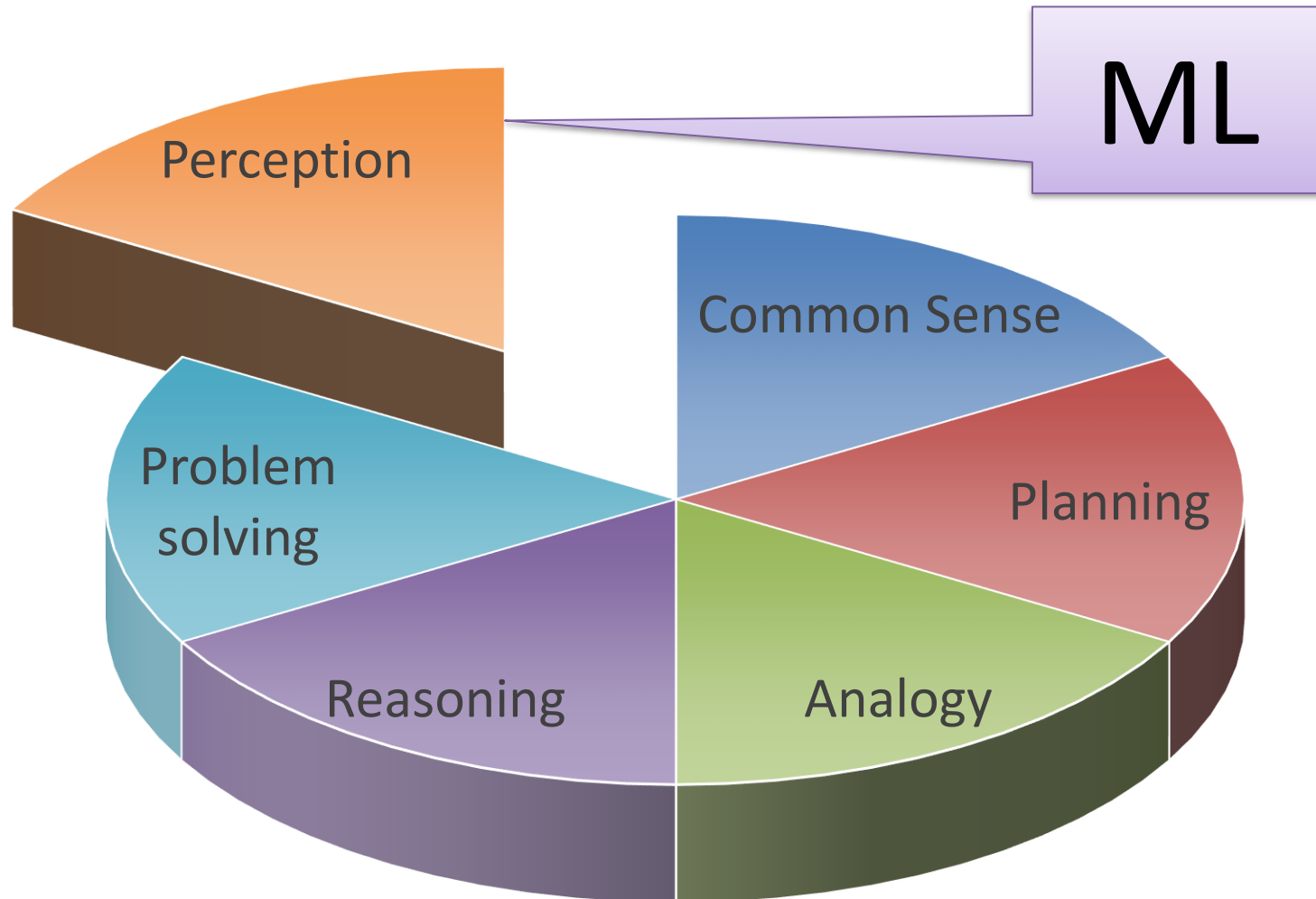
Thinking
slow

Thinking
fast

ML



Human intelligence = thinking fast & thinking slow



Thinking fast without thinking slow

Class: 793

Label: n04209133 (shower cap)

Certainty: 99.7%



& Weaknesses

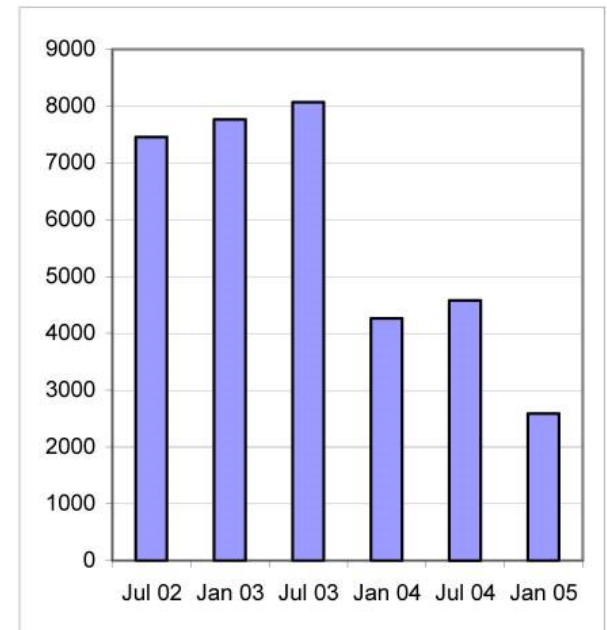
	Symbolic	Connectionist
Construction	Human effort	Data hunger
Scaleable	+/-	+/-
Explainable	+	-
Generalisable	Performance cliff	Performance cliff

Strengths & Weaknesses

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Construction	Human effort	Data hunger
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SNOMED CT
The global
language of
healthcare

300.00 definitions
40 years of effort,
10.000 updates every years



Strengths & Weaknesses

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☐ This one



☐ This one

10M training samples



4.8M training games

Strengths & Weaknesses

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worse with
more data



worse with
less data

Strengths & Weaknesses

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Rule: IF someone has dysphagia THEN that person may have esophageal cancer

someone

Variables

has dysphagia

Condition

that person may have
esophageal cancer

Conclusion or Action

Strengths & Weaknesses

	Symbolic	Connectionist
Construction	Human effort	Data hunger
Scaleable	+/-	+/-
Explainable	+	-
Generalisable	Performance cliff	Performance cliff



“panda”
55%

+



=



“gibbon”
99%



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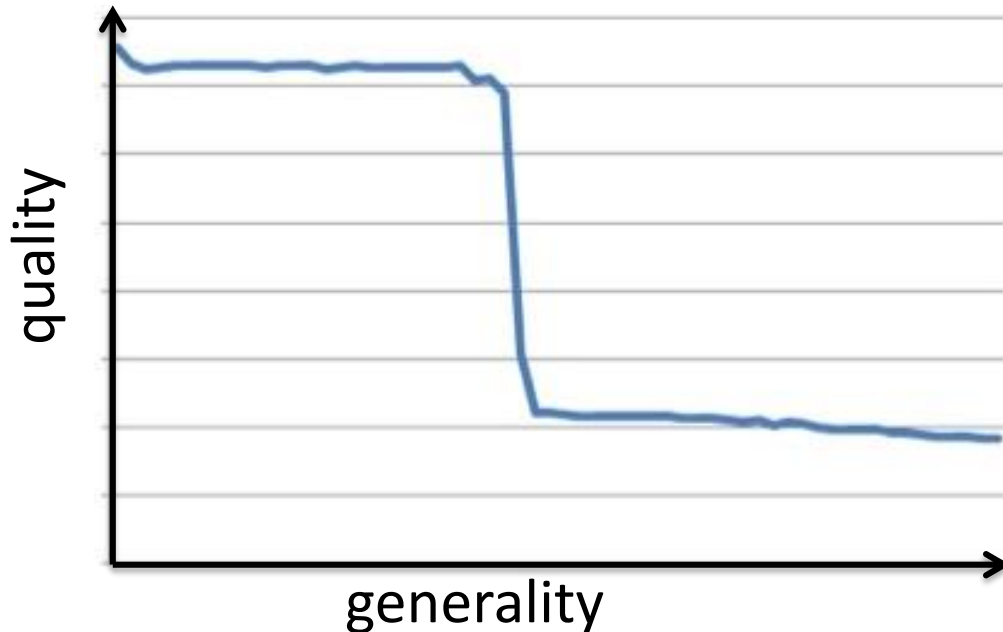


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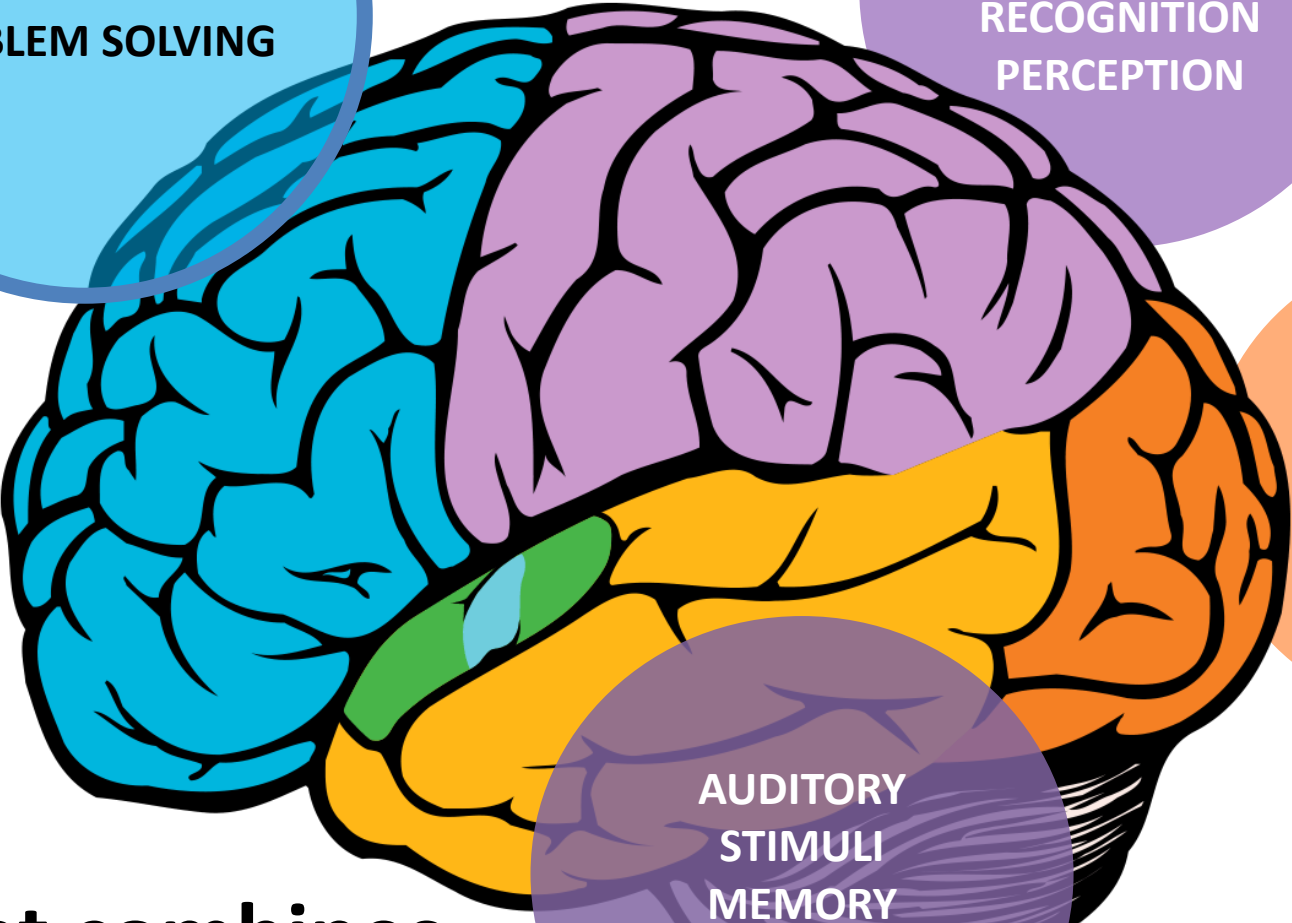
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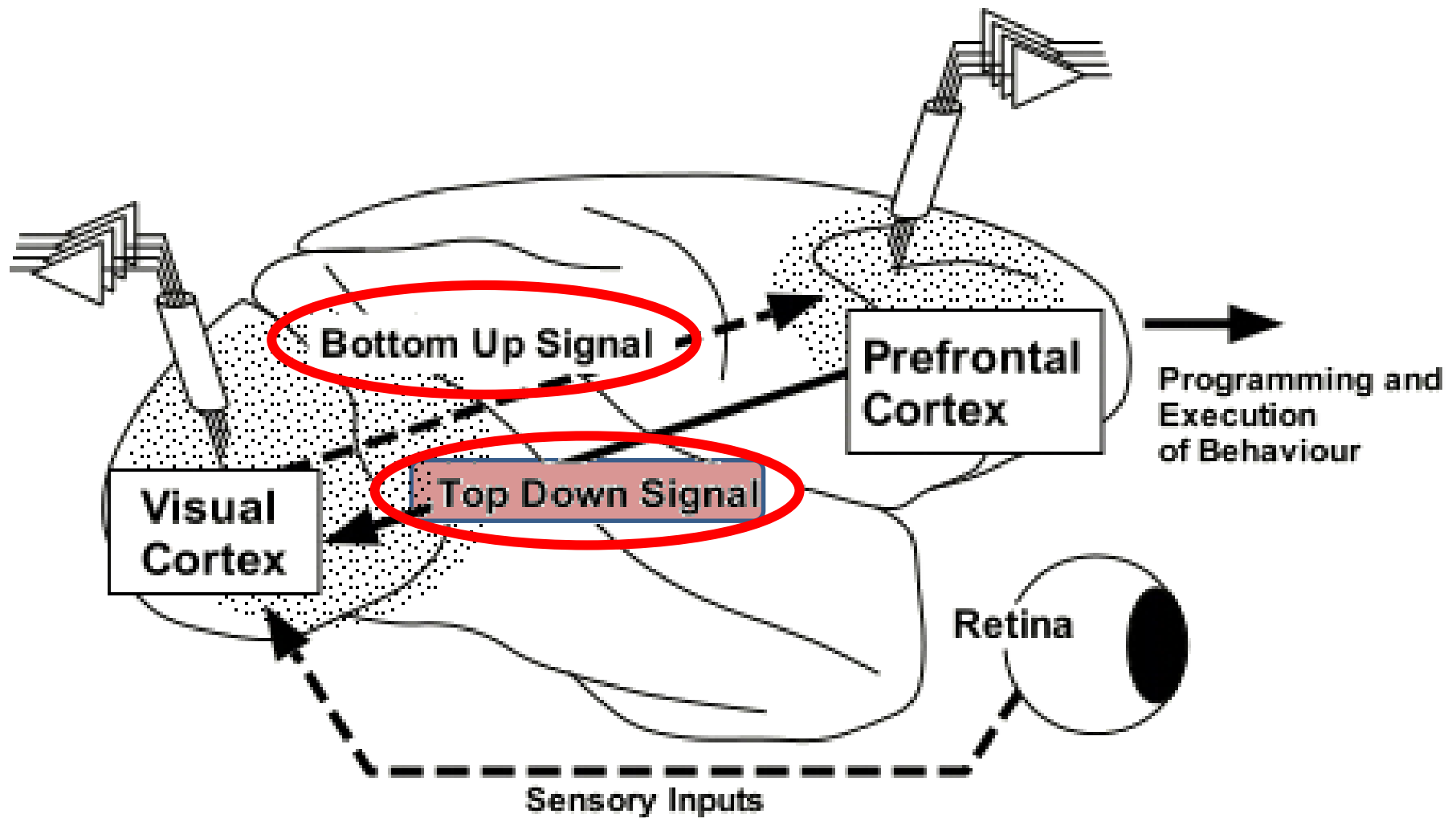
**REASONING
PLANNING
PROBLEM SOLVING**

**MOVEMENT
ORIENTATION
RECOGNITION
PERCEPTION**

**VISUAL
PROCESSING**

**AUDITORY
STIMULI
MEMORY
SPEECH**

**AI that combines
all the parts of our brain**



**AI that combines
all the parts of our brain**

AI that combines all the parts of our brain

Learning (Fast)	Reasoning (Slow)
Better with more data	Worse with more data
Needs lots of data	Already works with very little data
Not explainable	Explainable
Robust against noise	Brittle with noise

So what is a
Knowledge Representation?

Handbook of Knowledge Representation (1000 pages, ToC alone is 14 pages)

- propositional logic & satisfiability solvers
- first order logic & resolution
- description logic
- constraint (logic) programming
- nonmonotonic reasoning
- belief revision
- qualitative reasoning
- model-based diagnosis
- bayesian networks
- temporal logic
- spatial reasoning
- epistemic logic
- deontic logic
- situation calculus
- default logic
- event calculus
-



KR = Many Flavours

- **Boolean**

True: p ,

False: NOT p

Conditional: $p \rightarrow q$

- **Temporal**

NOW old $\rightarrow$ ALWAYS old

NOW young $\rightarrow$ SOMETIME not young

- **Epistemic**

truthful agents: $\text{KNOW } p \rightarrow p$

omniscient agents: $p \rightarrow \text{KNOW } p$

reflective agents: $\text{KNOW } p \rightarrow \text{KNOW KNOW } p$

- **Deontic**

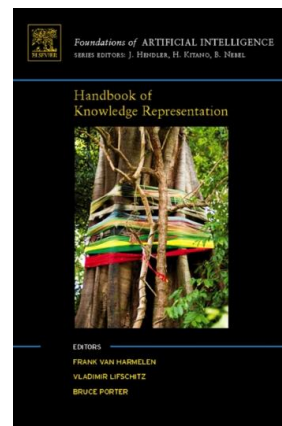
MAY $p \rightarrow$ not MUST not p

MUST $p \rightarrow$ not MAY not p

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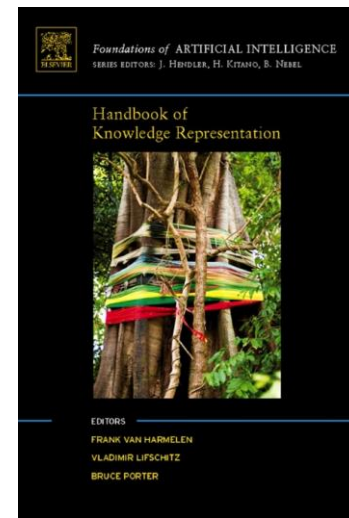
- **propositional logic & satisfiability solvers**
very simple language (hence very large problem descriptions), only for finite problems, but works surprisingly well.
- **first order logic & resolution**
compact representation, more expressive, hard to be efficient ("no free lunch" theorem ; law of conservation of misery)
- **description logic**
describe ontologies, class hierarchies of types, things and properties
- **constraint (logic) programming**
state only the requirements and impossibilities, let the machine figure out a feasible solution.
- **knowledge graphs**; relations between things as edges between nodes in a graph.
Very powerful when combined with webtechnology,
- **nonmonotonic reasoning**
sometimes when we know more, we can conclude less (surprise)
- **belief revision**
how to update knowledge in the light of new facts?
- **qualitative reasoning**
how to reason about the physical world without numbers?



Handbook of Knowledge Representation

(1000 pages, ToC alone is 14 pages)

- **model-based diagnosis**
compare predicted behaviour with observed behaviour and explain the differences.
- **bayesian networks**
combine logical knowledge with probabilistic knowledge
- **temporal logic**
how to reason about time
- **spatial reasoning**
how to reason about space
- **epistemic logic**: reasoning about knowledge and beliefs of other agents
(what they know, what they know that you know, etc).
- **deontic logic**
reasoning about permissions and obligations
- **situation calculus**
reasoning about (minimal) change in the world
- **default logic**
reasoning about (minimal) change in the world
- **event calculus**
reasoning about (minimal) change in the world

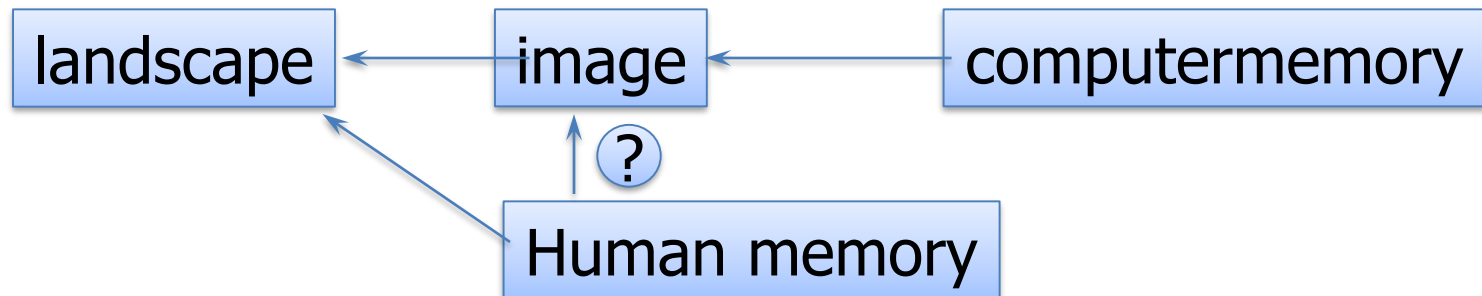


Kennisrepresentation = making a model

- AI = reasoning about the world through
a model of the world

Representation of a thing $\neq$ the thing itself

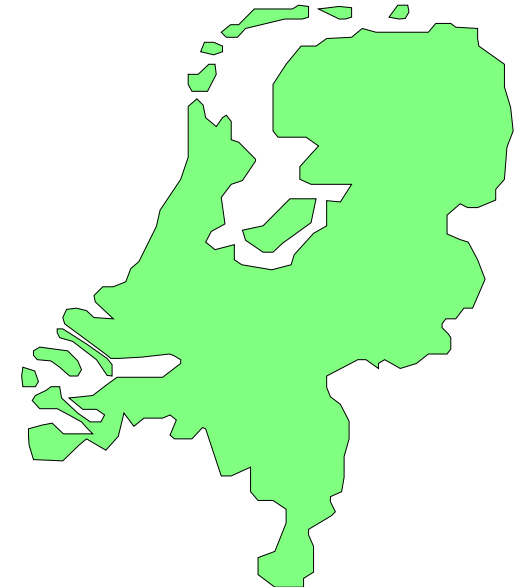
- Different representations of the same thing:
 π , 3.1415927, $11000103\frac{1}{7}$, $3\frac{99}{100}$,
- Representations of representations :



Knowledge Representation:

desirable properties

- **Abstraction**
Not more information than required
- **Precision**
Not less information than required
- **Efficiency**
store, retrieve, update,...



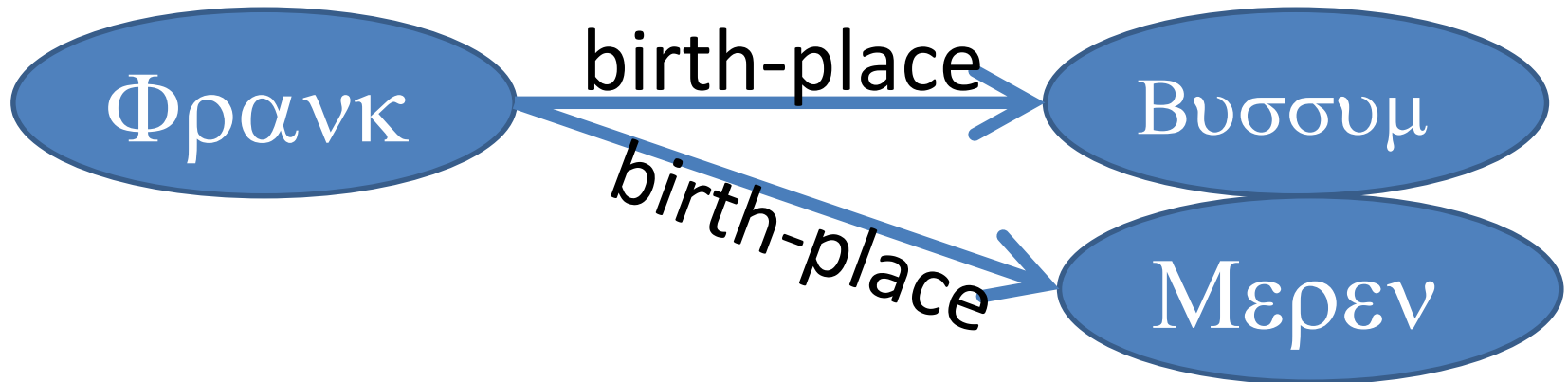
Knowledge Representation: desirable properties (2)

- Qualitative knowledge
≠ numeric, quantitative
- New knowledge must be derivable from old
≠ databases
- Distinguish
 - specific cases (facts)
 - General knowledge (rules, variables)
≠ databases
- Sufficiently rich structure
≠ databases

Knowledge Representation: desirable properties (2)

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explicit & formal semantics



- Φρανκ is person
- birth-place relates person to location

lowerbound

- birth-place relates 1 person to 1 location
- Βυσσυμ = Μερεν

upperbound

Semantics = predictable inference

Intensional vs. extensional



- **Extensional** = definition by enumeration
Vb: $\text{even-nr}(X) \leftrightarrow X = 0, 2, 4, 6, \dots$
- **Intensional** = definition by properties
Vb: $\text{even-nr}(X) \leftrightarrow \text{whole-nr}(X) \ \& \ \text{whole-nr}(X/2)$
- Extensional $\approx$ databases
= only finite (small!) domains
- Intensional: often difficult/impossible
Ex: “chair”?

